Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Identifies the ± sign (or just the negative) as the error. May be seen in any of the last five lines. May be indicated within Matthew's solution or described in words. Ignore other 'errors' identified. Condone identifying any of the last five lines as containing the error. PI	AO2.3	B1	$\pm\sqrt{y^2+1} = e^x - y$
	Gives a correct reason, referring to either e^x (or e^y), or the operand of a logarithm, being positive. Do not award if more than one error identified.	AO2.4	E1	It is an error because $y - \sqrt{y^2+1} < 0$ and $e^x > 0$ so there is a contradiction.
(b)	States the correct solution of the equation. Accept 10.0 [1787493] or $\frac{1}{2}(e^3 - e^{-3})$ ISW	AO1.1b	B1	arsinh $x = 3$ $x = \sinh 3$
				Total 3 marks

Q2.

(a) $x = \tanh y = \frac{e^y - e^{-y}}{e^y + e^{-y}}$

$xe^y + xe^{-y} = e^y - e^{-y}$

or $xe^{2y} + x = e^{2y} - 1$

M1

$\Rightarrow (x + 1)e^{-y} = e^y(1 - x)$

$\Rightarrow (x + 1) = e^{2y}(1 - x)$

A1

$e^{2y} = \frac{1+x}{1-x} \Rightarrow y = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$

AG

A1cso

(b) $y = \frac{1}{2}\ln(1+x) - \frac{1}{2}\ln(1-x)$

M1

$\frac{dy}{dx} = \frac{1}{2(1+x)} + \frac{1}{2(1-x)}$

A1

$= \frac{1-x+1+x}{2(1+x)(1-x)} = \frac{2}{2(1-x^2)} = \frac{1}{1-x^2}$

AG

A1cso

Alternative 1

$\frac{dy}{dx} = \frac{1}{2} \times \frac{(1-x)}{(1+x)} \times \frac{d}{dx}\left(\frac{1+x}{1-x}\right)$ *M1*

$\frac{dy}{dx} = \frac{1}{2} \times \frac{(1-x)}{(1+x)} \times \frac{(1-x)+(1+x)}{(1-x)^2}$ *A1*

$\Rightarrow \frac{dy}{dx} = \frac{1}{1-x^2}$ *A1 cso*

3

(c) $\int 4x \tanh^{-1} x \, dx = 4x \tanh^{-1} x - \int \frac{4x}{1-x^2} dx$

M1

$4x \tanh^{-1} x + 2\ln(1-x^2)$

A1

$\tanh^{-1}\frac{1}{2} = \frac{1}{2}\ln 3$

must simplify logarithm to ln 3

B1

Value of integral = $\ln 3 + 2\ln\frac{3}{4}$

any correct form

A1

$\ln\left(\frac{3^3}{2^4}\right)$

all working must be correct

A1cso

5

[11]

Q3.

(a) (i) $\frac{d}{du}\left(2u\sqrt{1+4u^2}\right) = \frac{8u^2}{\sqrt{1+4u^2}} + 2\sqrt{1+4u^2}$

M1 for clear use of product rule (condone one error in one term)

M1

correct unsimplified

A1

$\frac{d}{du}\left(\sinh^{-1} 2u\right) = \frac{2}{\sqrt{1+4u^2}}$

B1

$\frac{8u^2+2}{\sqrt{1+4u^2}} = \frac{2(1+4u^2)}{\sqrt{1+4u^2}} = 2\sqrt{1+4u^2}$

be convinced − must see this line OE

$\frac{d}{du}\left(2u\sqrt{1+4u^2} + 4\sinh^{-1} 2u\right) = 4\sqrt{1+4u^2}$

all working must be correct (not enough to just say k = 4)

A1cso

4

(ii) $\frac{1}{\text{“their”}k}\left[2u\sqrt{1+4u^2} + \sinh^{-1} 2u\right]_0^1$

anti differentiation

M1

$= \frac{\sqrt{5}}{2} + \frac{1}{4}\sinh^{-1} 2$

FT “their” k or even use of k

A1√

2

(b) (i) $y = \frac{1}{2}\cos 4x$ and $\frac{dy}{dx} = A\sin 4x$

$\frac{dy}{dx} = -2\sin 4x$

substituted into $\int K\, y\left(1+\left(\frac{dy}{dx}\right)^2\right)$ (dx)

clear attempt to use formula for CSA

M1

$(S =)\ \int_0^{\frac{\pi}{8}} 2\pi \times \frac{1}{2}\cos 4x\sqrt{1+4\sin^2 4x}\,dx$ = printed answer (combining $2 \times \frac{1}{2}$)

AG $\frac{dy}{dx} = -2\sin 4x$ ***and*** $2 \times \frac{1}{2}$ ***and*** *dx must be seen to award A1cso*

A1cso

2

(ii) $u = \sin 4x \Rightarrow du = 4\cos 4x \, dx$

condone $du = B\cos 4x \, dx$ *for M1*

M1

$(S =) \quad \frac{\pi}{4}\int_0^1 \sqrt{1+4u^2} \, (du)$

condone limits seen later

A1

use of their result from (a)(ii) ***correctly*** *FT "their"* B

m1

$(S =) \quad \frac{\pi\sqrt{5}}{8} + \frac{\pi}{16}\sinh^{-1} 2$

OE

A1cso

4

[12]

Q4.

(a) Sketch of $y = \cosh x$

approximately correct with minimum point above the x-axis, symmetrical about y-axis

B1

1

(b) Attempt to factorise

or complete square or use (correct unsimplified) formula

M1

$(3\cosh x - 5)(2\cosh x + 1) = 0$

A1

$\cosh x \neq -\frac{1}{2}$

indicated or stated (not merely neglected)

E1

$x = \ln\left(\frac{5}{3} + \sqrt{\frac{25}{9} - 1}\right)$

evidence of use of formula. Must see −1 or equivalent

M1

$= \pm \ln 3$

ft incorrect factorisation

A1F

A1 for ±

A1F

6

Alternative:

$3\left(\frac{e^x + e^{-x}}{2}\right) = 5$

$3e^{2x} - 10e^x + 3 = 0$

(M1)

$(3e^x - 1)(e^x - 3) = 0$

Correct factors

(A1F)

$x = \ln\frac{1}{3}$ or $\ln 3$

for both

(A1F)

NB if $\cosh x = \frac{e^x + e^{-x}}{2}$ used initially, M0 until quartic in e^x is factorised

M1 for $e^x - 3$ is a factor A1 if correct
M1 for $3e^x - 1$ is a factor A1 if correct
A1 for $x = \pm \ln 3$
E1 for showing remaining quadratic has no real roots

[7]

Q5.

(a) $\frac{(e^x + e^{-x})}{2}\frac{(e^y + e^{-y})}{2} - \frac{(e^x - e^{-x})}{2}\frac{(e^y - e^{-y})}{2}$

M0 if sinh and cosh confused
M1 for formula quoted correctly

M1A1

Correct expansions

Use of e^{xy} A0

A1

$$= \frac{1}{2}\left(e^{x-y} + e^{-(x-y)}\right) = \cosh(x-y)$$

AG

A1

4

(b) (i) $\cosh(x - \ln 2) = \cosh x \cosh(\ln 2) - \sinh x \sinh(\ln 2)$

M1

$$\left.\begin{aligned}\cosh(\ln 2) &= \frac{5}{4}\\ \sinh(\ln 2) &= \frac{3}{4}\end{aligned}\right\}$$ any method

Both correct

B1

$$\frac{5}{4}\cosh x = \frac{7}{4}\sinh x$$

A1F

$$\tanh x = \frac{5}{7}$$

AG

A1

4

Alternative:

$$\frac{e^{x-\ln 2} + e^{-x+\ln 2}}{2} = \frac{e^x - e^{-x}}{2}$$ *M1*

$$e^{x-\ln 2} = \frac{e^x}{2} \text{ or } e^{-x+\ln 2} = 2e^{-1}$$ $e^{-x+\ln 2} = 2e^{-1}$

used B1

$e^x = \sqrt{6}$ *A1*

$\tanh x = \frac{5}{7}$ *A1*

(ii) $$x = \frac{1}{2}\ln\left(\frac{1+\frac{5}{7}}{1-\frac{5}{7}}\right) \text{ or } \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{5}{7}$$

Could be embedded in (b)(i)

M1

$= \frac{1}{2}\ln 6$

A1

2

[10]

Q6.

(a) LHS = $\frac{1}{4}\left(e^{x}+e^{-x}\right)^{2}-\frac{1}{4}\left(e^{x}-e^{-x}\right)^{2}$

M1

Correct expansion of either square

A1

Shown equal to 1

AG

A1

3

(b) (i) 8 $\cosh^2 x - 3$

B1

1

(ii) Sketch of $y = \cosh x$

Must cross y-axis above x-axis

B1

1

(iii) $\cosh x = (\pm)1.25$

OE; ft errors in (b) (i); allow ± missing

B1F

$x = \ln\left(1.25 + \sqrt{1.25^2 - 1}\right)$

M1

= ln 2

A1F

$\ln\frac{1}{2}$ by symmetry

Accept – ln 2 written straight down
Alternatively, if solved by using
$e^{2x} - 2.5e^{x} + 1 = 0$, allow M1 for

$$x = \ln\left(\frac{2.5 \pm \sqrt{2.5^2 - 4}}{2}\right)$$

A1F

4

[9]

Q7.

(a) $\frac{d}{dx}\left(\cosh^{-1}\frac{1}{x}\right) = \frac{1}{\sqrt{\frac{1}{x^2} - 1}}\left(-\frac{1}{x^2}\right)$

M0 if $\frac{dy}{dx} = f(y)$ *and no attempt to substitute back to x*

M1A1

$$= \frac{-1}{x\sqrt{1 - x^2}}$$

AG

A1

3

(b) (i) $\frac{d}{dx}\left(\sqrt{1 - x^2}\right) = \frac{-2x}{2\sqrt{1 - x^2}}$

For numerator

B1

For denominator (not $(1 - x^2)^{\frac{1}{2}}$*)*

B1

$$\frac{dy}{dx} = \frac{-x}{\sqrt{1 - x^2}} + \frac{1}{x\sqrt{1 - x^2}}$$

$$= \frac{1 - x^2}{x\sqrt{1 - x^2}} = \frac{\sqrt{1 - x^2}}{x}$$

For attempt to put over a common denominator

M1

AG

A1

4

(ii) $s = \int_{\frac{1}{4}}^{\frac{3}{4}} \sqrt{1 + \frac{1 - x^2}{x^2}}\, dx = \int_{\frac{1}{4}}^{\frac{3}{4}} \frac{1}{x}\, dx$

For use of $\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

M1A1A1

$= \left[\ln x\right]_{\frac{1}{4}}^{\frac{3}{4}}$

M1

$= \ln \frac{3}{4} - \ln \frac{1}{4} = \ln 3$

AG

A1

5

[12]

Q8.

(a) $5\left(\frac{e^x - e^{-x}}{2}\right) + \left(\frac{e^x + e^{-x}}{2}\right)$

M0 if no 2s in denominator

M1

$= 3e^x - 2e^{-x}$

A1

2

(b) $3e^x - 2e^{-x} + 5 = 0$

$3e^{2x} + 5e^x - 2 = 0$

ft if 2s missing in (a)

M1

$(3e^x - 1)(e^x + 2) = 0$

A1F

$e^x \neq -2$

*any **indication** of rejection*

E1

$e^x = \frac{1}{3}$ $x = \ln \frac{1}{3}$

provided quadratic factorises into real factors

A1F

4

[6]

Examiner reports

Q1.

Many students failed to identify the error in the algebraic steps given in part (a). Of those who realised that ± was not acceptable, only half could explain why. A significant number of students suggested that completing the square was an invalid approach. Many others questioned the swapping of the variables x and y, with some explaining that variables could not be swapped. Some students decided that the variables had been swapped at the wrong point in the solution.

When answering part (b), many students failed to notice the connection with part (a). Of those who did spot the link, some decided that x was equal to $\sinh^{-1} 3$ instead of sinh 3. However, a number of students were able to solve the equation by algebraic means, and the correct answer sometimes followed several lines of skilful algebra.

Q2.

(a) Most students used $x = \tanh y$ and the given exponential form to obtain an expression in terms of x for e^{2y} and hence the printed result. Quite a few students thought that $\tanh^{-1}y = \frac{\sinh^{-1}y}{\cosh^{-1}y}$ and made no progress. Those students who started with $\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ were also rarely successful.

(b) The majority of students used the chain rule to differentiate the result from part (a). Those with greater insight used the law of logarithms to obtain the difference of two logarithms and the new form was then much easier to differentiate. Poor algebraic manipulation meant that some were forced to fudge their working to obtain the printed answer.

(c) This part of the question was the best source of marks for most students, though some were unable to integrate $\frac{x}{1-x^2}$ correctly. Those whose calculus was sound had no trouble expressing the final answer in the given form.

Q3.

(a) (i) The differentiation caused more difficulties than had been anticipated. Many students did not use the product rule on the first expression and those who did rarely had both terms correct. Even the derivative of $\sinh^{-1} 2u$ was done badly with many omitting the “2” in the numerator.

(ii) Those who did not obtain $k = 4$ were still able to score full marks here if they used their value of k or even “k” in their anti-differentiation.

(b) (i) Students needed to show the correct value of $\frac{dy}{dx}$ before substituting into the formula for the curved surface area in order to score full marks. Despite the printed answer, quite a few omitted the limits or the dx or failed to show the product $2 \times \frac{1}{2}$ in their working and lost an easy accuracy mark.

(ii) Once again the integration by substitution caused problems for many who omitted the factor $\frac{\pi}{4}$ or who did not have the correct limits for u. Because of errors in finding the value of k in part (a)(i) only the best students scored full marks here. Some credit was given for using their previous results correctly even though students did not have the correct value of k.

Q4.

The sketch of $y = \cosh x$ was almost invariably answered correctly in part (a). The most common approach to part (b) was to solve the quadratic equation and then to either use the formula for $\cosh^{-1} x$ in its logarithmic form in which case the solution -ln 3 was generally lost, in spite of the hint given in part (a), or to express the equation $\cosh x = 5 / 3$ in terms of e^x in which case both answers were usually correctly obtained. Some students chose to express the quadratic equation in terms of e^x, but this led to a quartic in e^x, which they were then unable to solve. The reasons for rejecting the solution $\cosh x = –1 / 2$ were generally adequate.

Q5.

The bookwork in part (a) was generally well done, and errors if they did occur were almost always when candidates thought that, for instance, $e^x e^y$ was e^{xy}. In part (b)(i), those candidates who took the hint given in (a) were more successful than those who converted the given equation into exponential form, the latter being unable to handle expressions such as $e^{x - \ln 2}$. Part (b) was usually correct.

Q6.

Part (a) was generally well done apart from a few candidates who were unable to square $\frac{1}{2}(e^x - e^{-x})$ successfully.

Parts (b)(i) and (b)(ii) likewise were well done although $\pm\frac{1}{2}\pi$ did appear on the x-axis of some sketches in part (b)(i). However in part (b)(iii), those candidates using the logarithmic formula for$\cosh^{-1} x$ from the formulae booklet arrived at a single value of x, namely ln2, but failed to realise that the sketch in part (b)(ii) was intended to give them a hint that –ln 2 was also a solution of the equation. On the other hand, candidates who used the exponential form of either $\cosh x$ or $\cosh^2 x$ automatically produced both answers provided their working was correct, but some of these candidates were unable to handle the algebra leading to a quadratic equation in either e^x or e^{2x}.

Q7.

Few candidates were able to supply a correct proof of part (a). The derivative of $\cosh^{-1}\left(\frac{1}{x}\right)$ was almost invariably given as $\frac{1}{\sqrt{\left(\frac{1}{x}\right)^2 - 1}}$ (followed by the correct answer after a small amount of spurious algebra. Candidates fared better in part (b)(i), although it was a little surprising how many candidates were unable to differentiate $\sqrt{1 - x^2}$ correctly. Those candidates who continued and attempted part (b)(ii) frequently produced a correct solution. Solutions to this part which went awry were those in which candidates wrote $\sqrt{1+\left(\frac{\sqrt{1-x^2}}{x}\right)^2}$ followed by $\sqrt{1+\frac{1-x}{x}}$ (which incidentally gave the correct answer!), or in some cases where candidates arrived at $\sqrt{\frac{1}{x^2}}$ or $\sqrt{x^{-2}}$ but were unable to take the square root correctly.

Q8.

Most candidates scored full marks for this question. The only significant error was in the use of incorrect formulae for sinh x and cosh x. These formulae were sometimes quoted without the halves, ie sinh x was written as $e^x - e^{-x}$ and cosh x as $e^x + e^{-x}$. In part (b), a small number of candidates thought that $\ln(3e^{2x} + 5e^x - 2)$ was equal to $\ln 3e^{2x} + \ln 5e^x - \ln 2$.